

Rohit Menon

Robotics Researcher | Active Perception, Reasoning and Manipulation

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[Website](#) | [Google Scholar \(242 citations; h-index: 9; i10-index: 9\)](#) | [GitHub](#) | [LinkedIn](#)

Citation metrics as of October 2026

Research Profile

Robotics researcher and PhD candidate investigating **prior-guided active perception** for mapping, object search and manipulation in partially observed environments. Developed methods combining **shape completion, scene-graph reasoning, evidential uncertainty and geometric planning**, with validation on greenhouse and mobile-manipulation systems. Research experience spans **learning-based perception, neuro-evolution, shared autonomy and multi-arm manipulation**, complemented by industry experience in robot architecture and real-time control.

Education

PhD in Computer Science

Feb. 2022–Mar. 2027 (Expected)

[University of Bonn](#), Germany | Advisor: [Prof. Dr. Maren Bennewitz](#)

MSc in Automation and Robotics

Oct. 2012–Mar. 2015

[TU Dortmund University](#), Germany | Grade: 1.2; top 5% of cohort

Bachelor's in Electrical Engineering

Aug. 2004–June 2008

[SPCE](#), [University of Mumbai](#), India | Grade: 76%; silver medal

Research Experience

Research Associate / PhD Researcher

Feb. 2022–Present

[Humanoid Robots Lab](#), [University of Bonn](#), Germany; [PhenoRob Cluster of Excellence](#)

- Developed [SG-DOR](#) for **scene-graph reasoning with graph neural networks**, inferring plant-organ attachments and ranking direction-conditioned leaf occlusions for fruit intervention; equal-first author, IROS 2026.
- Developed [O-STaR](#) for **open-vocabulary object search**, grounding LLM location priors in scene geometry and updating probabilistic beliefs on dynamic scene graphs; GRC 2026, with an extended version under review at ICRA 2027.
- Developed [SG-AMP](#), integrating **uncertainty-aware depth completion, persistent panoptic mapping, plant scene graphs and semantics-aware view-motion planning**; preprint, 2026.
- Developed [NBV-SC](#) for **shape-completion-guided next-best-view planning**, targeting unseen fruit surfaces for mapping and reconstruction; validated with a UR5e in a commercial greenhouse; first author, IROS 2023.
- Developed [EvidMTL](#) and [EvidKimera](#) for **evidential depth and semantic prediction, uncertainty quantification and 3D semantic surface mapping**; evaluated on NYUv2 and zero-shot ScanNet mapping; first author, IROS 2025.
- Co-developed [HortiBot's three-arm harvesting system](#), integrating **active perception, online plant mapping and manipulation**; equal-contribution author, IROS 2024.
- Co-developed [DawnIK](#), an **optimization-based, collision-aware inverse-kinematics solver** for heterogeneous multi-arm systems; equal-contribution author, Humanoids 2023.
- Designed and procured research hardware for [HortiBot](#), a **Neobotix-based omnidirectional mobile manipulator**, and the **PhenoRob Phase 2 quadruped and sensors**; annotated tomato and pepper plant datasets and generated synthetic perception data.

- Solely responsible for **modelling, control and communication** of NeTTUN's tunnel-boring maintenance manipulator, including remote teleoperation.
- Independently developed **synchronized, multi-point, jerk-limited trajectory generation** for high-DoF systems with configurable kinematic constraints.
- Built an **autonomous unknown-object picking pipeline**, combining deep-learning grasping with optimization-based motion planning; investigated **tactile sensing and reactive grasping**.
- Developed **path planning and trajectory generation for mobile manipulation** in Hybr-iT, and **reactive grasping and upper-limb motion planning for the RH5 humanoid** in TransFit.
- Led **DFKI's iLEVATOR contribution**, managing budget, procurement and sponsor reporting; the mobile bin-picking system was **tested and deployed at the end user's factory**.

- Devised and implemented **adaptive shared autonomy** for assistive manipulation, combining automatic grasp selection from robot kinematics and user commands with **virtual-fixture trajectory guidance**.
- Preserved continuous user control of the object approach direction while guiding velocity commands toward a selected grasp pose.
- Validated the method in **full-dynamics simulation, a user study and on the real robot**; thesis grade: 1.3; resulting publication at ICRA 2016.

- Investigated **neuro-evolutionary learning of echo state networks** for **mobile robot navigation**.

Publications

Full author lists; **Rohit Menon** is highlighted; * denotes equal contribution. Published versions and their arXiv preprints are listed once.

Journal Articles

- [1] Rajasree Padmakumari Hemachandran Nair, **Rohit Menon**, Ralf Kemkemmer. [Generalised Image Processing Method for Quantitative Analysis of Nucleus, Cell and Focal Adhesion Clusters](#). *Current Directions in Biomedical Engineering*, 7(2), pp. 558–561, 2021.
- [2] Srinidhi Balasubramanian, Donald Michael McFarland, Sotiria Koloutsou-Vakakis, Kan Fu, **Rohit Menon**, Christopher Lehmann, Mark J. Rood. [Effect of Grid Resolution and Spatial Representation of NH₃ Emissions from Fertilizer Application on Predictions of NH₃ and PM_{2.5} Concentrations in the United States Corn Belt](#). *Environmental Research Communications*, 2(2), article 025001, 2020.

Conference Publications

- [3] **Rohit Menon***, Niklas Mueller-Goldingen*, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. [SG-DOR: Learning Scene Graphs with Direction-Conditioned Occlusion Reasoning for Pepper Plants](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. Accepted; first two authors contributed equally.
- [4] Benno Wingender*, Nils Dengler*, **Rohit Menon**, Sicong Pan, Maren Bennewitz. [A Robust Placeability Metric for Model-Free Unified Pick-and-Place Reasoning](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [5] **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz. [O-STaR: Open-Vocabulary Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs](#). *German Robotics Conference (GRC)*, 2026.

- [6] **Rohit Menon**, Nils Dengler, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. [EvidMTL: Evidential Multi-Task Learning for Uncertainty-Aware Semantic Surface Mapping from Monocular RGB Images](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [7] Allen Isaac Jose*, Sicong Pan*, Tobias Zaenker, **Rohit Menon**, Sebastian Houben, Maren Bennewitz. [GO-VMP: Global Optimization for View Motion Planning in Fruit Mapping](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [8] Ahmed Shokry, Walid Gomaa, Tobias Zaenker, Murad Dawood, **Rohit Menon**, Shady A. Maged, Mohammed I. Awad, Maren Bennewitz. [Context-Based Meta Reinforcement Learning for Robust and Adaptable Peg-in-Hole Assembly Tasks](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [9] Nils Dengler, Jesper Mücke, **Rohit Menon**, Maren Bennewitz. [Efficient Manipulation-Enhanced Semantic Mapping With Uncertainty-Informed Action Selection](#). *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2025.
- [10] Christian Lenz*, **Rohit Menon***, Michael Schreiber, Melvin Paul Jacob, Sven Behnke, Maren Bennewitz. [HortiBot: An Adaptive Multi-Arm System for Robotic Horticulture of Sweet Peppers](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- [11] Benedikt Kreis, Nils Dengler, Jorge de Heuvel, **Rohit Menon**, Hamsa Perur, Maren Bennewitz. [Compact Multi-Object Placement Using Adjacency-Aware Reinforcement Learning](#). *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2024.
- [12] Salih Marangoz*, **Rohit Menon***, Nils Dengler, Maren Bennewitz. [DawnIK: Decentralized Collision-Aware Inverse Kinematics Solver for Heterogeneous Multi-Arm Systems](#). *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2023.
- [13] **Rohit Menon**, Tobias Zaenker, Nils Dengler, Maren Bennewitz. [NBV-SC: Next Best View Planning Based on Shape Completion for Fruit Mapping and Reconstruction](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [14] Tobias Zaenker, Julius Rückin, **Rohit Menon**, Marija Popović, Maren Bennewitz. [Graph-based View Motion Planning for Fruit Detection](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [15] Nils Dengler, Sicong Pan, Vamsi Kalagaturu, **Rohit Menon**, Murad Dawood, Maren Bennewitz. [Viewpoint Push Planning for Mapping of Unknown Confined Spaces](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [16] Benedikt Kreis, **Rohit Menon**, Bharath Kumar Adinarayan, Jorge de Heuvel, Maren Bennewitz. [Reactive Correction of Object Placement Errors for Robotic Arrangement Tasks](#). *International Conference on Intelligent Autonomous Systems (IAS-18)*, pp. 257–270, 2023.
- [17] Salih Marangoz, Tobias Zaenker, **Rohit Menon**, Maren Bennewitz. [Fruit Mapping with Shape Completion for Autonomous Crop Monitoring](#). *IEEE International Conference on Automation Science and Engineering (CASE)*, 2022.
- [18] Jörn Vogel, Katharina Hertkorn, **Rohit U. Menon**, Máximo A. Roa. [Flexible, Semi-Autonomous Grasping for Assistive Robotics](#). *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4872–4879, 2016.

Workshop Papers and Posters

- [19] Benno Wingender, Nils Dengler, **Rohit Menon**, Sicong Pan, Maren Bennewitz. [Robust Placeability Estimation for Model-Free Unified Pick-and-Place Reasoning](#). *ICRA Workshop on Manipulation Robustness: Towards Human-Level Robustness under Real-World Challenges*, 2026.
- [20] **Rohit Menon**, Yasmin Schmiede, Maren Bennewitz, Hermann Blum. [Open-Vocabulary and Semantic-Aware Reasoning for Search and Retrieval of Objects in Dynamic and Concealed Spaces](#). *IROS Workshop on Perception and Planning for Mobile Manipulation in Changing Environments*, 2025. *Best Poster Award*.
- [21] Nils Dengler, Sicong Pan, Vamsi Kalagaturu, **Rohit Menon**, Murad Dawood, Maren Bennewitz. [Learning Interactive Viewpoint Planning for Foresighted Object Discovery Considering Human Preferences](#). *ICRA Workshop on Human-Aligned Reinforcement Learning for Autonomous Agents and Robots*, 2024.
- [22] José de Gea Fernández, Dennis Mronga, Martin Günther, Sebastian Stock, Nils Niemann, Hendrik Wiese, **Rohit Menon**, Elsa Andrea Kirchner, Stefan Stiene. [Towards Contextual Robots for Collaborative Manufacturing](#). *IROS Workshop on Human-Robot Cooperation and Collaboration in Manipulation: Advancements and Challenges*, 2018. *Poster*.

Preprints and Manuscripts under Review

- [23] **Rohit Menon**, Shiva Rudra Lolla, Niklas Mueller-Goldingen, Gokul Chenchani, Ribana Roscher, Maren Bennewitz. [SG-AMP: Scene-Graph-Guided Active Perception and Semantics-Aware Motion Planning for Pepper Plants](#). *arXiv:2609.01579*, 2026. *Preprint*.
- [24] **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz. O-STaR: Effective Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs. *Manuscript*, 2026. *Extended version submitted to ICRA 2027; under review*.
- [25] Sicong Pan, Luca Lobefaro, Moein Taherkhani, Xuying Huang, **Rohit Menon**, Cyrill Stachniss, Maren Bennewitz. [Efficient View Planning Guided by Previous-Session Reconstruction for Repeated Plant Monitoring](#). *arXiv:2510.07028*, 2025. *Preprint; revised in 2026; submitted for review*.

Teaching and Supervision

MSc Theses

- Supervised **five MSc theses** at the University of Bonn, yielding **two full conference papers and one workshop paper**; two supervised theses received awards in 2026.
- Supervised research on **active perception, fruit mapping and manipulation**.
- At DFKI, supervised an MSc thesis on **multi-sensor fusion for in-hand object localization**, slip compensation and object placement accuracy.

PhD Mentoring

- Supervising **three junior PhD students** on interactive perception and manipulation at the Humanoid Robots Lab.

MSc Lab Projects

- Supervised robotics lab projects at the University of Bonn on perception, mapping and manipulation.

Seminars

- Supervised the [Humanoid Robots seminar](#) in summer 2026 on sampling-based motion planning with scene graphs under perception constraints, including research-proposal and presentation preparation.

Humanoid Robotics Lecture

Summer 2025, 2026

- Redesigned **manipulation and perception lectures**, exercises and assessments; integrated modern perception pipelines and semantic mapping.
- Conducted oral and written examinations and managed grading.

Cognitive Robotics Lecture

2022, 2023, 2024

- Served as **stand-in lecturer and teaching assistant**, delivering probabilistic-robotics sessions and designing assignments and examinations.
- Supervised practical programming assignments and managed grading.

Earlier Teaching

- Graded **Reinforcement Learning** tutorials at the University of Bremen during the DFKI appointment.
- Organized a one-day **MATLAB workshop** for undergraduate engineering students at SPCE in 2008.

Academic Service

- **Journal reviewer**: IEEE Robotics and Automation Letters (RA-L); The International Journal of Robotics Research (IJRR).
- **Conference reviewer**: ICRA, IROS, CASE, Humanoids; Robotics: Science and Systems (RSS), 2026.
- Served on the **PhD interview committee** at the Humanoid Robots Lab, University of Bonn.
- Member of the **Robotics Institute Germany technical cluster on semantic perception**.
- Organizing the **PhenoRob Study Group on Photosynthesis**.

Research Funding and Project Development

- Assisted in writing the [Agriloop](#) proposal (EUR 400,000) for **foundation model adaptation for human-in-the-loop robotic intervention in greenhouses**.
- Contributed to the **PhenoRob Phase 2 HRL PhD** proposal.

Invited Talks

Active Perception for Robotics in Agriculture

May 2025

Pint of Science Festival, Bonn, Germany; public-facing talk on active perception for crop monitoring and harvesting.

Active Perception

Dec. 2022

Indian Institute of Information Technology (IIIT), Kottayam, India.

Outreach and Public Engagement

- Featured in a [WDR Bonn television interview](#), demonstrating HortiBot and discussing robotic horticulture.
- Demonstrated **HortiBot** during the [visit of NRW State Premier Hendrik Wüst](#) to the University of Bonn, February 2024.
- Presented **active perception and fruit mapping** at the [ICRA@40 live demonstration](#), Rotterdam, 2024.
- Conducted robot demonstrations at **Tag der offenen Tür (2024, 2025)**, [Wissenschaftsfest am Hofgarten \(2024\)](#) and **Bonn Nacht der Technik (2023)**.
- Set up and demonstrated the **Hybr-iT mobile manipulator at Hannover Messe 2019** while at DFKI.
- Demonstrated Neura robots at **GITEX 2021**, Dubai, with coverage in [The National](#).

Awards, Scholarships and Honours

Best Poster Award

2025

IROs Workshop on Perception and Planning for Mobile Manipulation in Changing Environments.

Deutschlandstipendium

2013

Scholarship for academic excellence, TU Dortmund University.

Siemens Excellence in Commissioning Trophy

2011

Siemens India Conference.

Silver Medal, Bachelor's in Electrical Engineering

2008

Sardar Patel College of Engineering, University of Mumbai.

Ratan Tata Scholarship

2006, 2007

Scholarship for excellence in engineering during the bachelor's degree.

Best Student in Science Faculty

2004

Bhavan's Junior College, Mumbai; all-round performance among more than 1,000 students.

First Rank, Class XII HSC Examination

2004

Bhavan's Junior College, Mumbai.

Second Rank, Class X SSC Examination

2002

IES Secondary School, Mumbai.

Industry Experience

Founding Engineer / Group Leader, AI for Manipulation

Nov. 2019–Jan. 2022

Neura Robotics GmbH, Metzingen, Germany

- **Independently built MAiRA's perception–manipulation architecture** as the AI team's first software engineer, including C++/ROS libraries, device drivers and communication interfaces.
- Rewrote **Xenomai/Orocos RTT tasks for hard real-time control and actuator communication** via SOEM EtherCAT.
- Developed selective pick-and-place using semantic segmentation and depth fusion; integrated **voice- and gesture-based manipulation commands**.
- Built **human collision avoidance using RGB, depth and radar fusion** with real-time 3D human pose tracking; developed a robot tic-tac-toe demonstrator.
- Promoted to **AI for Manipulation group leader** in September 2021, with **hiring responsibility**.

Engineering and Commissioning / Senior Executive

Sept. 2008–Sept. 2012

Siemens Ltd, Thane, India

- Developed **AC/DC drive controls and PLC/HMI software**; commissioned machinery, tuned control loops and optimized machine logic.
- Managed technical interfaces with clients, consultants and vendors; progressed from graduate trainee to executive and senior executive.

Research Methods and Technical Skills

Methods: Active perception; scene graphs; graph neural networks; shape completion; evidential deep learning; uncertainty quantification; semantic mapping; open-vocabulary mobile manipulation; shared autonomy; neuro-evolution and echo state networks.

Programming and learning: C++, Python, MATLAB/Simulink, PyTorch, PyTorch Geometric.

Robotics and simulation: ROS, ROS 2, Isaac Sim, Gazebo, Blender.

Control and libraries: Xenomai, Orocos RTT, SOEM EtherCAT, OpenCV, PCL, Eigen, NLOpt.

Languages: English C1 (TOEFL: 116/120), German B1, Hindi, Malayalam.